

SOFTWARE ENGINEERING PROJECT

Smart Home Automation System

Field	Details
Group Number	Group 03
Case Study	Smart Home Automation System
Course	Software Engineering — Winter 2026
Instructor	Dr. Samer Elkababji

Team Members

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System Overview

The Smart Home Automation System is an IoT-based platform that integrates and controls smart home devices including thermostats, lighting, cameras, and alarms. It supports remote device control, automation routines, real-time alerts, and AI-driven optimization through mobile and web applications.

Tools Used

Tool	Purpose
PlantUML	All UML and C4 diagram generation
Visual Studio Code	Report and source file editing
GitHub	Version control and collaboration

Repository Structure

/ README.md | docs/report.md | docs/report.pdf | uml/*.puml | uml/*.png

Member Contributions

Member	Role	Commits
Abdulrhman Atassi	Interaction Modeler — Use case diagrams and descriptions	18
Sarah Shobaki	System Architect — C4 diagrams, container design	32
Osama Zaid	Sequence Modeler — Sequence diagrams	1
Mariam Laith	Behavior Modeler — Activity diagrams	1

1. System Description

The Smart Home Automation System is an IoT-based platform designed to integrate and centrally control smart home devices. It serves three actors: a Primary Homeowner with full control, Family Members with scoped access, and a System Administrator for backend operations.

The system exposes functionality through mobile and web applications. An AI/ML module learns per-user behavior patterns to suggest and automate routines. A dedicated IoT Gateway translates between HTTP/REST and physical device protocols (Zigbee, MQTT, Bluetooth). A Redis cache provides real-time device state reads. All persistent data lives in a central PostgreSQL database.

Four core operational flows drive the design: device control, autonomous automation triggered by sensor events, user and device onboarding, and safety-critical incident detection and response.

2. Part I — Context

2.1 C4 Level 1 — System Context Diagram

Presents the system as a black box showing who uses it and what external systems it depends on. Three actors identified: Primary Homeowner (full control), Family Member (scoped permissions), and System Administrator (direct backend access). Five external systems: Mobile/Web App, IoT Devices, Notification Service, AI/ML Module, and Authentication Service (OAuth 2.0).

Design decision: Guest User excluded — the system is private to registered household members only. AI/ML Module appears external at L1 because the system is treated as a black box at this level; it moves inside the boundary at L2.

[C4 Level 1 — System Context | [uml/c4_l1_context.png](#)]

See /uml/ folder for diagram image

2.2 C4 Level 2 — Container Diagram

Opens the black box to reveal 11 deployable containers and their communication protocols.

Container	Type	Key Rationale
Mobile App / Web App	Client	Explicitly required by case study
API Gateway	REST Entry Point	Centralizes auth, rate limiting, and routing for all requests
Device Management Service	Backend	Core service for device state, registration, and command dispatch
Automation Engine	Backend	Isolated from Device Management — rule evaluation is different from device I/O
User & Preference Service	Backend	Per-user profiles, roles, permissions, and household management
Notification Service	Backend	Internal dispatch before handing off to external push/email
IoT Gateway	Protocol Bridge	Translates HTTP/REST to Zigbee/MQTT/Bluetooth — mandatory layer
AI/ML Service	ML Module	Integrated module per case study — inside system boundary at L2
Central Database	PostgreSQL	Persistent storage for all entities
Device State Cache	Redis	Sub-millisecond state reads — polling PostgreSQL directly is unsound for IoT

[C4 Level 2 — Container Diagram | [uml/c4_l2_container.png](#)]

See /uml/ folder for diagram image

2.3 Activity Diagrams

Six activity diagrams with swimlanes cover all actors, all containers, and all primary use cases. Each models a distinct process — no two diagrams cover the same flow.

ID	Flow	Why It Was Modeled	Key Swimlanes
AD 1	Device Control	Core real-time user flow — the most frequent system interaction	Homeowner/FM, App, API GW, Device Mgmt, IoT GW
AD 2	Automation & AI Trigger	Background intelligence — sensor-initiated, no human actor	IoT GW, Device Mgmt, Automation Engine, AI/ML, Notification
AD 3	User Registration & Onboarding	Foundational — system has no users or devices without this flow	New User, App, API GW, User Service, Device Mgmt, IoT GW
AD 4	Alert & Incident Response	Safety-critical path — the flow that must not fail	IoT Sensor, IoT GW, Device Mgmt, Automation Engine, Notification, HO/FM, AI
AD 5	Manage Family Members	Non-trivial — involves cross-service permission propagation to Device Management	Homeowner, App, API GW, User Service, Device Mgmt, Notification, New FM
AD 6	Admin Monitoring	Different access pattern — no IoT layer, direct backend access	Admin, Web App, API GW, Device Mgmt, User Service, Automation Engine

[AD1 — Device Control | [uml/ad1_device_control.png](#)]

See /uml/ folder for diagram image

[AD2 — Automation & AI | [uml/ad2_automation_ai.png](#)]

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[AD3 — Onboarding | [uml/ad3_onboarding.png](#)]

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[AD4 — Alert Response | [uml/ad4_alert_response.png](#)]

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[AD5 — Manage Family Members | [uml/ad5_manage_family.png](#)]

See /uml/ folder for diagram image

[AD6 — Admin Monitoring | [uml/ad6_admin_monitoring.png](#)]

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3. Part II — Interactions

3.1 Use Case Diagrams

Two diagrams provided. Split by actor type to maintain readability. Actor-to-package associations used throughout — documented design decision to avoid unreadable individual lines with 17 use cases and 5 actors.

Diagram	Actors	Packages Covered
UC1 — User Interactions	Homeowner, Family Member	Account, Devices, Automation, Monitoring, Preferences
UC2 — System & Admin	System Administrator, Automation Engine, AI/ML Service	Administration, Automation & AI, Notifications

Relationships: <<include>> used for Log In — mandatory prerequisite for all protected use cases. <<extend>> used for Trigger Alert (extends Control Device) and Generate Suggestion (extends Create Automation Rule) — both are conditional optional extensions.

[UC1 — User Interactions | [uml/uc1_user_interactions.png](#)]
See /uml/ folder for diagram image

[UC2 — System and Admin | [uml/uc2_system_admin.png](#)]
See /uml/ folder for diagram image

3.2 Use Case Descriptions

Eight tabular descriptions covering one use case per major functional area.

UC-01: Log In

Use Case ID	UC-01
Actors	Homeowner, Family Member
Description	A registered user authenticates to gain access to permitted features
Preconditions	Registered and verified account exists
Postconditions	User authenticated, JWT token issued, redirected to dashboard
Main Flow	1. Open app → 2. Enter credentials → 3. API Gateway forwards to Auth Service → 4. JWT returned → 5. Access granted based on role
Alternate Flow	Invalid credentials → error + retry. Unverified account → prompt verification
Exceptions	5 failed attempts → account locked
Includes / Extends	— / —

UC-02: Control Device

Use Case ID	UC-02
Actors	Homeowner, Family Member
Description	Authenticated user sends a control command to a registered smart device
Preconditions	Logged in. Device registered and online. User has permission
Postconditions	Device executes command. State updated in cache and DB. User notified
Main Flow	1. Select device → 2. Issue command → 3. API GW validates auth + permission → 4. Device Mgmt forwards to IoT GW → 5. IoT GW dispatches via Zigbee/MQTT → 6. Device ACKs → 7. State updated → 8. Confirmation sent
Alternate Flow	No ACK → retry up to 3 times → all fail → report failure + trigger alert
Exceptions	Device offline → 503. Permission denied → 403
Includes / Extends	UC-01 Log In / UC-06 Trigger Alert

UC-03: Add / Remove Device

Use Case ID	UC-03
Actors	Homeowner
Description	Homeowner registers a new smart device or removes an existing one
Preconditions	Logged in as Homeowner. Device powered and within range
Postconditions	Device registered in DB. IoT GW establishes communication channel
Main Flow	1. Navigate to devices → 2. Scan QR / enter ID → 3. System validates → 4. IoT GW performs handshake → 5. Status set to Online → 6. Assign name and room
Alternate Flow	Duplicate ID → conflict error. Handshake fails → Unreachable status
Exceptions	Invalid ID → rejected
Includes / Extends	UC-01 Log In / —

UC-04: Create Automation Rule

Use Case ID	UC-04
Actors	Homeowner
Description	Homeowner defines a trigger-action rule for automatic device control
Preconditions	Logged in as Homeowner. At least one device registered
Postconditions	Rule saved to DB. Automation Engine begins evaluating it
Main Flow	1. Navigate to automation → 2. Define trigger + action → 3. Save → 4. Automation Engine registers rule
Alternate Flow	Rule conflict → warn user and request confirmation
Exceptions	Invalid configuration → validation error
Includes / Extends	UC-01 Log In / UC-08 Generate AI Suggestion

UC-05: Manage Family Members

Use Case ID	UC-05
Actors	Homeowner
Description	Homeowner adds/removes household members and assigns device permissions
Preconditions	Logged in as Homeowner
Postconditions	Member account created/updated. Permissions propagated to Device Management Service
Main Flow	1. Add member → 2. Enter email + permissions → 3. Invitation sent → 4. Member accepts → 5. Permissions propagated to Device Mgmt ACL
Alternate Flow	Email exists → link existing account. Invitation expired after 48h
Exceptions	Household member limit reached → rejected
Includes / Extends	UC-01 Log In / —

UC-06: Trigger Alert

Use Case ID	UC-06
Actors	Automation Engine
Description	System detects anomalous event and dispatches high-priority alert to homeowner
Preconditions	Sensor event or device failure has occurred
Postconditions	Alert delivered via push/email. Incident logged to DB
Main Flow	1. Anomaly detected → 2. Severity classified → 3. Notification Service dispatches push + email → 4. Incident logged
Alternate Flow	Push fails → fallback to email only
Exceptions	Notification Service unavailable → queue for retry
Includes / Extends	— / UC-02 Control Device

UC-07: Execute Automation Rule

Use Case ID	UC-07
Actors	Automation Engine
Description	Automation Engine evaluates rules against device state changes and executes matches
Preconditions	Active rules exist. Device state change received
Postconditions	Matched actions executed. Execution logged
Main Flow	1. State change received → 2. Load active rules → 3. Evaluate → 4. Execute matches → 5. Log execution
Alternate Flow	No match → log and no action
Exceptions	Rule references deleted device → skip and flag
Includes / Extends	— / —

UC-08: Generate Automation Suggestion

Use Case ID	UC-08
Actors	AI / ML Service
Description	AI analyzes usage history to generate personalized automation suggestions
Preconditions	Sufficient usage history in DB
Postconditions	Suggestion auto-applied (if enabled) or sent to homeowner for review
Main Flow	1. Read usage history → 2. Identify pattern → 3. Generate suggestion → 4a. Auto-apply if enabled OR 4b. Send for user review
Alternate Flow	No pattern → schedule next analysis
Exceptions	Insufficient data → skip until threshold met
Includes / Extends	— / UC-04 Create Automation Rule

3.3 Sequence Diagrams

Seven sequence diagrams — one high-level stakeholder view and six detailed developer diagrams, one per activity diagram. Each detailed SD traces directly to its AD for full vertical traceability.

ID	View	Flow Modeled	Traces To
SD 1	High-Level (Stakeholder)	Device Control — plain language, 5 participants	AD1, UC-02
SD 2	Detailed (Developer)	Device Control — JWT, Redis, Zigbee, HTTP codes, DB writes	AD1, UC-02
SD 3	Detailed (Developer)	Automation & AI — sensor-initiated, rule evaluation, AI suggestion lifecycle	AD2, UC-07, UC-08
SD 4	Detailed (Developer)	Registration & Onboarding — form to email verify to IoT handshake	AD3, UC-01, UC-03
SD 5	Detailed (Developer)	Alert Response — emergency rules, false alarm, AI recalibration	AD4, UC-06
SD 6	Detailed (Developer)	Manage Family — permission propagation across Device Mgmt ACL	AD5, UC-05
SD 7	Detailed (Developer)	Admin Monitoring — health aggregation, device removal, user deactivation	AD6

[SD1 — High-Level Device Control | [uml/sd1_highlevel.png](#)]

See /uml/ folder for diagram image

[SD2 — Detailed Device Control | [uml/sd2_device_control_detailed.png](#)]

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[SD3 — Automation & AI | [uml/sd3_automation_detailed.png](#)]

See /uml/ folder for diagram image

[SD4 — Onboarding | [uml/sd4_onboarding_detailed.png](#)]

See /uml/ folder for diagram image

[*SD5 — Alert Response* | *uml/sd5_alert_detailed.png*]

See /uml/ folder for diagram image

[*SD6 — Family Member* | *uml/sd6_family_member_detailed.png*]

See /uml/ folder for diagram image

[*SD7 — Admin Monitoring* | *uml/sd7_admin_detailed.png*]

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4. Part III — Structure

4.1 Class Diagram

16 domain classes derived directly from entities that appeared in the sequence and activity diagrams. Every class is traceable to at least one SD or AD.

Relationship Type	Relationships
Generalization (Inheritance)	User (abstract) < -- Homeowner, FamilyMember, Administrator
Composition (5)	Household *-- Device, User *-- Preference, Device *-- DeviceState, AutomationRule *-- AutomationLog, FamilyMember *-- Permission
Aggregation (1)	Household o-- User (users exist independently of households)
Association (8)	AutomationRule→Device, Incident→Device, Notification→User, AIModel→AISuggestion, AISuggestion→AutomationRule, Permission→Device, Homeowner→Invitation, Invitation→FamilyMember

Key decisions: User is abstract — no plain User is ever instantiated. Composition for Household-Device — devices cannot exist without a household. Aggregation for Household-User — users exist before joining a household. AISuggestion is separate from AutomationRule — it has its own lifecycle and only becomes a rule after user acceptance.

[[Class Diagram — Smart Home Automation System](#) | [uml/class_diagram.png](#)]

See /uml/ folder for diagram image

5. Part IV — Behavior

The system is classified as hybrid — it exhibits both event-driven behavior (IoT sensor events trigger device state transitions and autonomous automation) and data-driven behavior (user data, device telemetry, and usage history are processed and transformed into outputs). Accordingly two behavioral models are provided.

5.1 State Diagram — Device Lifecycle

Models all states a Device can be in and the events that cause transitions. Device is the most architecturally significant entity — it appears in every operational flow.

State	Description
Unregistered	Device exists physically but not yet added to the system
Initializing	Onboarding in progress — IoT Gateway handshake underway
Online (composite)	Connected and registered. Contains Idle and Executing substates
Idle	Online, no active command — ready to receive
Executing	Actively processing a dispatched command
Unreachable	Registered but not responding to communication
Alert	Anomaly detected — Automation Engine executes emergency rules on entry
Maintenance	Flagged for inspection — all rules targeting this device disabled on entry
Removed	Deregistered — terminal state, no further transitions

Key decisions: Online is a composite state because Idle and Executing only exist in the context of a connected device. Alert does not transition directly to Removed — a safety-critical system must route through Maintenance to enforce human inspection before removal.

[State Diagram — Device Lifecycle | [uml/state_diagram_device.png](#)]

See /uml/ folder for diagram image

State-Stimulus Table

Current State	Stimulus / Event	Next State	Action Taken
Unregistered	Homeowner registers device ID	Initializing	IoT GW initiates handshake
Initializing	Handshake successful	Online	Status set Online in DB + cache
Initializing	Handshake failed	Unreachable	Status set Unreachable, user notified
Online / Idle	Command dispatched	Online / Executing	Device Mgmt forwards to IoT GW
Online / Executing	ACK received	Online / Idle	State updated in cache + DB, user notified
Online / Executing	All retries exhausted	Unreachable	Failure reported, alert triggered
Online	Sensor anomaly detected	Alert	Emergency rules executed, alert dispatched

Current State	Stimulus / Event	Next State	Action Taken
Online	Deregistered by admin/homeowner	Removed	Record soft-deleted, rules disabled
Alert	Incident resolved	Online	Incident marked resolved, AI feedback logged
Alert	Marked as false alarm	Online	False alarm logged, AI threshold recalibrated
Alert	Alert count exceeds threshold	Maintenance	Rules disabled, admin notified
Unreachable	Heartbeat received	Online	Status updated in cache + DB
Unreachable	Offline > 24 hours	Maintenance	Device flagged, admin notified
Unreachable	Admin force removes	Removed	Record soft-deleted, rules disabled
Maintenance	Admin clears device	Online	Status restored, rules re-enabled
Removed	—	—	Terminal — no transitions

5.2 Data Flow Diagram

Models the data-driven dimension only. Event-driven flows (commands, automation triggers, incidents) are excluded — they are already covered by the State Diagram and Activity Diagrams. The DFD focuses on the data management pipeline.

DFD Level 0 — Context

Shows the data pipeline as a single process with three external entities: Homeowner/Family Member (provides registration data, preferences; receives AI suggestions and reports), IoT Devices (provides telemetry; receives config sync), and System Administrator (receives health reports and analytics).

[DFD Level 0 — Data Management Context | [uml/dfd_l0_context.png](#)]
See /uml/ folder for diagram image

DFD Level 1 — Data Management Pipeline

Process	Data Stores Used	Description
P1 — Manage User & Household Data	D1 Users & Preferences	Registration, profile updates, preferences, family member management
P2 — Manage Device & Configuration Data	D2 Devices & Configurations	Device registration, telemetry storage, configuration sync, firmware tracking
P3 — AI Learning Data Pipeline	D3 Usage History, D4 AI Model	Reads usage history, trains model, generates and stores automation suggestions
P4 — Generate Reports & Dashboards	D1, D2, D3, D4, D5 Reports	Aggregates all data stores into dashboards for users and health reports for admin

Note: Device State Cache (Redis) is intentionally excluded from the DFD — it is a real-time infrastructure component supporting the event-driven layer, not the data management pipeline. It is correctly modeled in C4 L2 and the sequence diagrams.

[DFD Level 1 — Data Management Pipeline | [uml/dfd_l1_pipeline.png](#)]
See /uml/ folder for diagram image

6. Design Decisions and Assumptions

#	Decision	Justification
1	Guest User excluded	System is private to registered household members only
2	Family Member as distinct actor from Homeowner	Case study mentions per-user preferences — implies individual profiles with different permission scopes
3	AI/ML external at C4 L1, internal at C4 L2	At L1 system is a black box. At L2 case study describes AI as an integrated module
4	IoT Gateway as separate container	Physical devices speak Zigbee/MQTT; backend speaks HTTP/REST — protocol translation is a distinct concern
5	Redis cache for device state	Sub-millisecond reads required for real-time dashboards. Polling PostgreSQL directly is architecturally unsound for IoT
6	Actor-to-package associations in UC diagrams	17 use cases + 5 actors = unreadable individual lines. Package-level associations are a documented professional practice
7	6 activity diagrams instead of 1	Full actor coverage requires separate diagrams for Family Member (AD5) and System Administrator (AD6)
8	System classified as hybrid	Significant event-driven behavior (IoT events, state transitions) AND data-driven behavior (AI pipeline, reporting)
9	DFD scoped to data management only	Event-driven flows already covered by State Diagram and Activity Diagrams — mixing them would produce an inaccurate diagram
10	User is abstract	No plain User is ever instantiated — every person is a Homeowner, FamilyMember, or Administrator
11	AI Suggestion separate from AutomationRule	Suggestion has its own lifecycle and ID — becomes a rule only after user acceptance

7. Traceability Matrix

Use Case → Activity Diagram → Sequence Diagram → Class

Use Case	Activity Diagram	Sequence Diagram	Class(es)
UC-01 Log In	AD3	SD4	User hierarchy
UC-02 Control Device	AD1	SD1, SD2	Device, DeviceState
UC-03 Add/Remove Device	AD3	SD4	Device, Household
UC-04 Create Automation Rule	AD2	SD3	AutomationRule
UC-05 Manage Family Members	AD5	SD6	FamilyMember, Permission, Invitation
UC-06 Trigger Alert	AD4	SD5	Incident, Notification
UC-07 Execute Automation Rule	AD2	SD3	AutomationRule, AutomationLog
UC-08 Generate AI Suggestion	AD2	SD3	AIModel, AISuggestion

Container Coverage

Container	Activity Diagrams	Sequence Diagrams
Mobile / Web App	AD1,3,5,6	SD1,2,4,5,6,7
API Gateway	AD1,3,5,6	SD2,4,5,6,7
User & Preference Service	AD3,5	SD4,6,7
Device Management Service	AD1,2,3,4,5	SD2,3,4,5,6,7
Automation Engine	AD2,4,6	SD2,3,5,7
Notification Service	AD2,3,4,5	SD2,3,4,5,6,7
IoT Gateway	AD1,2,3,4	SD2,3,4,5
AI/ML Service	AD2,4	SD3,5
Central Database	All ADs	All SDs
Device State Cache	AD1,2,4	SD2,3,5

Behavior Coverage

Diagram	Dimension	Traces To
State Diagram — Device Lifecycle	Event-driven	AD1, AD4, SD2, SD5
DFD Level 0	Data-driven context	C4 L1
DFD Level 1	Data-driven pipeline	AD3, AD5, SD4, SD6, SD7